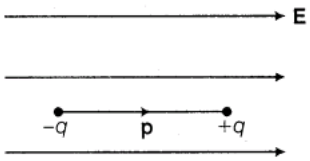
QPT-7 Solution

1. **Ans: (c)** Zero and minimum

   Solution:

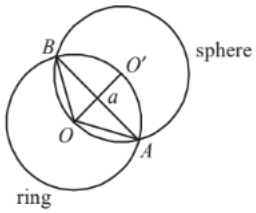


   The total force on dipole is zero because $F = qE$ is applied on each charge but in opposite direction. The potential energy is $\mathrm{U} = -\vec{\mathrm{p}} \cdot \vec{\mathrm{E}}$, which is minimum when $\vec{\mathrm{p}}$ and $\vec{\mathrm{E}}$ are parallel.

2. **Ans: (d)** Low melting point, high specific resistance.

   Solution: Fuse wire should be such that it melts immediatley when strong current flows through the circuit. The same is possible if its melting point is low and resistivity is high.

3. **Ans: (c)** $\frac{q}{3\varepsilon_0}$



   Solution:
   Charge on ring q, centre of ring $= O$
   Centre of sphere $= O'$
   Linear charge density of ring, $\lambda = \frac{q}{2\pi a}$
   Charge on arc AB of ring,

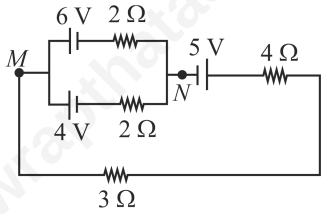
$$q_{AB} = \lambda(\text{arc}\, AB) = \frac{1}{2\pi a} \cdot q \cdot \frac{2\pi a}{3}$$

$$q_{AB} = q/3$$

   i.e., charge enclosed by sphere $= q/3$
   $\therefore$ Flux coming out of sphere $= q/3\varepsilon_0$.

4. **Ans: (c)** $(1/2)$ A

   Solution:



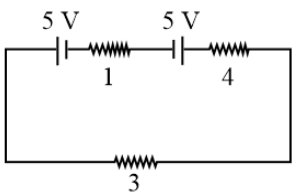
   Equivalent EMF between points M and N,

$$\varepsilon_{\text{eq}} = \frac{\frac{\varepsilon_1}{R_1} + \frac{\varepsilon_2}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{\frac{6}{2} + \frac{4}{2}}{\frac{1}{2} + \frac{1}{2}}$$

$$\varepsilon_{\text{eq}} = \frac{3+2}{1} = 5\ \text{V}$$

   Equivalent resistance between M and N,

$$\frac{1}{R} = \frac{1}{2} + \frac{1}{2} \Rightarrow R = 1\ \Omega$$



   Equivalent emf of circuit,

$$\varepsilon_{\text{eq}} = 5 - 5 = 0$$

   So, there will be no current in main circuit.

   Current will only flow in small loop in the upper circuit.

u/wrathgrass

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$6 - 4 - 4i = 0$

Hence, current through $6\ \text{V}$ battery is $\frac{1}{2}\ \text{A}$.

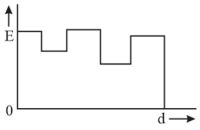
5. **Ans: (d)** If both assertion and reason are false.

Solution: The potential energy stored between the plates of capacitor,
$U = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}\frac{Q^2}{\varepsilon_0 A}x \quad \left(\because C = \frac{\varepsilon_0 A}{x}\right)$
Force $F = \frac{-\partial U}{\partial x} = -\frac{\partial}{\partial x}\left(\frac{1}{2}\frac{Q^2 x}{\varepsilon_0 A}\right) = -\frac{1}{2}\frac{Q^2}{\varepsilon_0 A}$ Hence, assertion is wrong as force is proportional to the square of charge.
Negative sign shows force is attractive.
From this, reason is also incorrect.

6. **Ans: (c)** 100

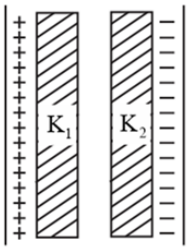
Solution: Voltage across X is $2v$
So $\left(\frac{E_1}{500+X}\right)X = 2$
$\Rightarrow \left(\frac{12}{500+X}\right)X = 2$
$\Rightarrow 12X = 1000 + 2X$
$\Rightarrow 10X = 1000$
$X = 100\Omega$

7. **Ans: (c)**



Solution:

Here we have to calculate the graph between E and d in the given situation



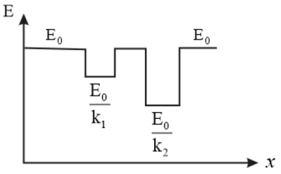
We know that in dielectric E becomes E/k

where

E = Electric fjeld

k = Dielectric constants

Since $(K_1 < K_2)$, The third option will be correct.



8. **Ans: (a)** $\frac{1}{2}\left(\frac{q}{\varepsilon_0} - \phi\right)$

Solution: According to Gauss law, $\phi_{\text{total}} = \frac{q}{\varepsilon_0}$
Let electric flux linked with surfaces A, B and C are $\phi_{\text{A}}, \phi_{\text{B}}$ and ϕ_{C} respectively. That is
$\phi_{\text{total}} = \phi_{\text{A}} + \phi_{\text{B}} + \phi_{\text{C}}$
Since $\phi_{\text{C}} = \phi_{\text{A}}$
$\therefore 2\phi_{\text{A}} + \phi_{\text{B}} = \phi_{\text{total}} = \frac{q}{\varepsilon_0}$
or $\phi_A = \frac{1}{2}\left(\frac{q}{\varepsilon_0} - \phi_B\right)$
But $\phi_B = \phi \quad (\text{given})$
Hence, $\phi_A = \frac{1}{2}\left(\frac{q}{\varepsilon_0} - \phi\right)$

9. **Ans: (a)** $\frac{2\times10^6}{\ln 2}\ \Omega$

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Solution: As $Q = Q_0 e^{-t/\tau}$. When energy is 50% then $Q = \frac{Q_0}{\sqrt{2}}$

$$\frac{Q_0}{\sqrt{2}} = Q_0 e^{-t/\tau}$$

$$e^{t/\tau} = \sqrt{2}$$

$$\frac{t}{\tau} = \ln(\sqrt{2}) \quad \tau = \frac{t}{\ln(\sqrt{2})}$$

$$RC = \frac{2}{\ln 2}$$

$$R = \frac{2}{C \ln 2} = \frac{2}{10^{-6} \times \ln 2} = \frac{2 \times 10^6}{\ln 2}\,\Omega$$

10. **Ans: (a)** $(\mathrm{A}) - (\mathrm{R}), (\mathrm{B}) - (\mathrm{S}), (\mathrm{C}) - (\mathrm{P}), (\mathrm{D}) - (\mathrm{Q})$

Solution:

Drift velocity, $V = \left(\frac{eE}{m}\right)\tau$

Electrical resistivity, $\rho = \frac{E}{J}$

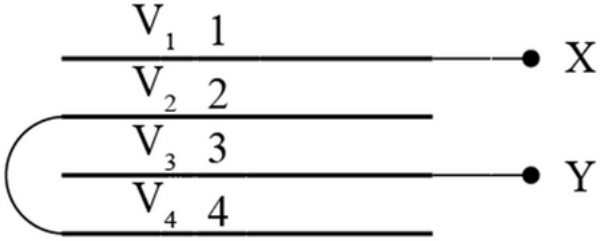
Relaxation period, $\tau = \frac{m}{ne^2\rho}$

Current Density, $J = neV_d$

$A - R,\ B - S,\ C - P,\ D - Q$

11. **Ans: (c)** $\frac{2}{3}\ \frac{\varepsilon_0 A}{d}$

Solution:



From the figure, it is clear that two capacitances will form the equivalent arrangement as shown in figure.
Let C be the capacitance of each capacitor.
The net capacitance of P and Q connected in parallel $C_{PQ} = 2C$
Now, C_{PQ} and C are in series.

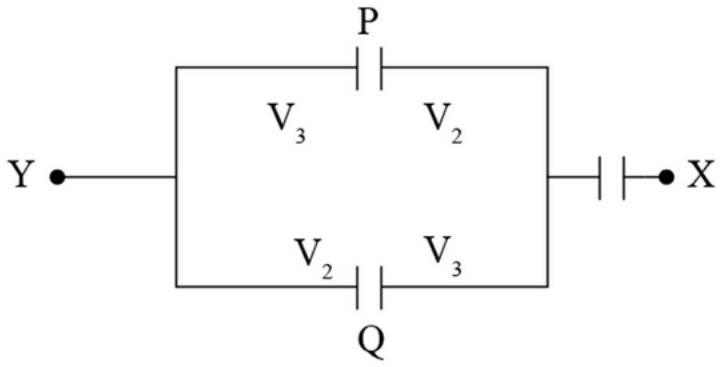
$$\therefore C_{XY} = \frac{C \cdot C_{PQ}}{C + C_{PQ}}$$

$$= \frac{C.2C}{C + 2C} = \frac{2}{3}\,C$$

The capacitance of parallel plate capacitor

$$C = \frac{\varepsilon_0 A}{d}$$

$$\Rightarrow C_{XY} = \frac{2}{3}\ \frac{\varepsilon_0 A}{d}$$



12. **Ans: (a)** $4.03 \times 10^{12}\ \mathrm{m}$

Solution: Step 1: Calculate the acceleration due to the electric field
The electric force F_e on the charged particle is given by:
$F_e = Q \times E$
The acceleration a_c due to the electric field is:
$a_c = \frac{F_e}{m} = \frac{Q \times E}{m} = \frac{Q}{m} \times E$
Substituting the given values:

$$a_e = \left(9.6 \times 10^7\right) \times \left(5 \times 10^5\right) \mathrm{m/s^2}$$

$$a_e = 4.8 \times 10^{13}\ \mathrm{m/s^2}$$

Step 2: Time to descend distance y
The time t taken to descend a vertical distance y under the influence of gravity is given by:

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$$y = \frac{1}{2} g t^2$$

$$t^2 = \frac{2y}{g} = \frac{2 \times 0.84\ \text{m}}{10\ \text{m/s}^2} = 0.168\ \text{s}^2$$

$$t = \sqrt{0.168} \approx 0.41\ \text{s}$$

Step 3: Calculate the horizontal displacement

The horizontal displacement x is given by:

$x = \frac{1}{2} a_n t^2$

Substituting the values:

$$x = \frac{1}{2} \times 4.8 \times 10^{13} \times 0.168\ \text{m}$$

,

$$x = 4.03 \times 10^{12} \text{ml}$$

13. **Ans: (c)** 60000Ω in series

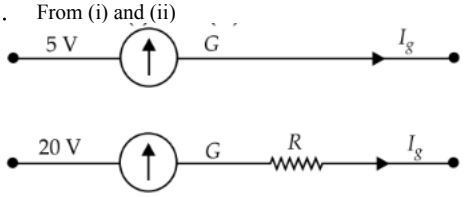
Solution: For a voltmeter, in which current I_g passes,

when range is lower, $5 = G \times I_g$ ...(i)

when range is higher, $20 = (G + R)\, I_g$ ...(ii)

where G = resistance of voltmeter $= 20000\Omega$

$\therefore$ From (i) and (ii)

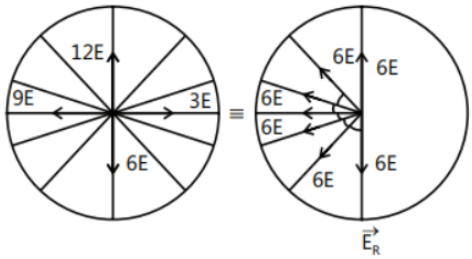


$\frac{5}{20} = \frac{G}{G+R}$ or $\frac{1}{4} = \frac{G}{G+R}$

or $R = 3G = 3 \times 20000 = 60000$ or extra resistance $= 60000\Omega$ in series ,

14. **Ans: (b)** $9:30$

Solution:



Electric field due to various charges is shown in the diagram.

So, on solving, net electric field will be along the line as shown in the given diagram.

The charges at $8:00, 10:00$ will be at angle 30° below and above the horizontal line and at $7:00, 11:00$ will be at angle 60° below and above the horizontal line.

On resolving the components along x & y $-$axes,

$$\tan\theta = \frac{P}{B}$$

$$\tan\theta = \frac{6E}{6E + 2 \times 6E\cos(30^\circ) + 2 \times 6E\cos(60^\circ)}$$

$$\tan\theta = \frac{6E}{6E + 6E + 6E\sqrt{3}}$$

$$\tan\theta = \frac{1}{2 + \sqrt{3}}$$

$$\tan\theta = 2 - \sqrt{3}$$

Hence, $\theta = 15^\circ$.

So, the net electric field will be above $9:00$ at angle 15°.

The angle between two consecutive numbers in a clock is 30°.

So, the direction will be in the middle of $9:00$ & $10:00$.

Hence, at $9:30$.

;

15. **Ans: (b)** $\frac{105}{11}$ A

Solution: For cells in parallel $\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} = \frac{1}{1} + \frac{1}{0.6}$

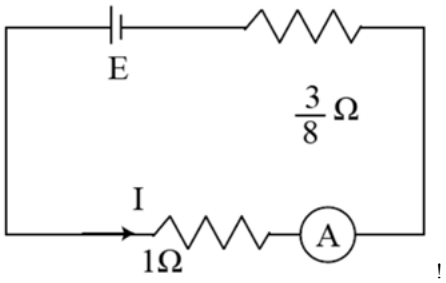
$\frac{1}{r} = 1 + \frac{5}{3} = \frac{8}{3}$

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$r = \frac{3}{8}\,\Omega$

$\frac{E}{r} = \frac{E_1}{r_1} + \frac{E_2}{r_2} \Rightarrow \frac{8E}{3} = \frac{10}{1} + \frac{15}{0.6} \quad \Rightarrow \quad \frac{8E}{3} = 35 \Rightarrow E = \frac{105}{8}\,V$

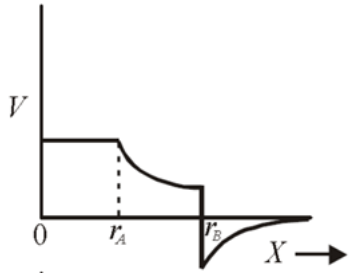
$I = \frac{E}{\frac{3}{8}+1} = \frac{105}{11}\,A$



!

16. **Ans: (d)** All of the above

Solution: Initially, $Q_1 = 24\mu C \quad [\because Q = CV]$
Finally, $V_{CQ} = \frac{24}{\frac{6}{5}+1} = \frac{24\times5}{11}$ V
$Q_3' = Q_2' = \frac{6}{5} \times \frac{24\times5}{11} = 13\mu$C (approx)
$Q_1' = 1 \times \frac{24\times5}{11} = 11\mu$C (approx)
$V_1' = \frac{Q'}{C_1} = 11$ V
$V_3' = \frac{Q_3'}{C_3} = \frac{13}{3} = 4.3$ V(approx) ^

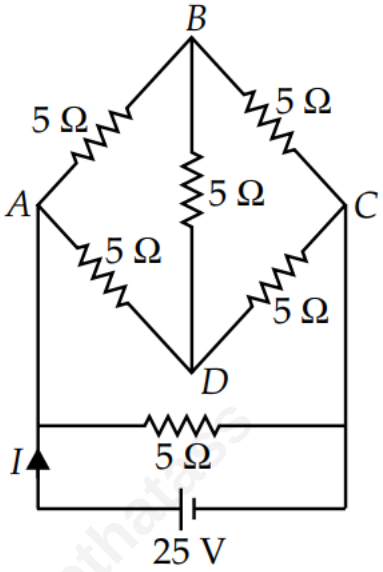


17. **Ans: (c)**

Solution: $(Q_B| > |Q_A|)$ (given). Inside the shell B, the potential is a constant and negative.
But V_A inside r_A is positive and greater than that of B because $r_A < r_B$. Upto r_A, one as hence positive potential. Beyond $r_A, V_A \propto (1/r)$, this is $> V_B$, one gets the course $\propto 1/r$. At B, the negative potential is more therefore the potential line is still negative. (c) explains all these factors. /

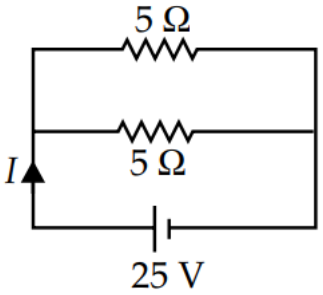
18. **Ans: (a)** 10 A

Solution: It is a balanced Wheatstone bridge. Hence no current flows through resistance of arm BDand the resistance of arm BD is ineffective



$\therefore \frac{R_{AB}}{R_{BC}} = \frac{R_{AD}}{R_{DC}}$
Resistance of arm
$ABC = 5\Omega + 5\Omega = 10\Omega$
Resistance of arm
$ADC = 5\Omega + 5\Omega = 10\Omega$
The effective resistance between A and C is
$\frac{1}{R_{AC}} = \frac{1}{10} + \frac{1}{10}$ or $R_{AC} = 5\Omega$
Now the equivalent circuit becomes
$\therefore$ The equivalent resistance of the circuit

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$\frac{1}{R} = \frac{1}{5} + \frac{1}{5}$ or $R = 2.5\Omega$
Current in the circuit,
$I = \frac{V}{R} = \frac{25}{2.5} = 10$ A !

19. **Ans: (d)** If both assertion and reason are false.

Solution: Only a charged particle initially at rest moves along the electric field line. In case, the charged particle is in motion and its initial velocity is at an angle with the electric field line, the particle may follow a parabolic path. ;

20. **Ans: (a)** $\frac{1}{4\pi\varepsilon_0} \cdot \frac{Q^2}{4R}$

Solution: The capacitances of two are $C_1 = 4\pi\varepsilon_0 R$ and $C_2 = 4\pi\varepsilon_0(2R)$
Initial energy $= E_i = \frac{Q^2}{2C_1}$
Final energy $= E_f = \frac{Q^2}{2C_2}$
Heat produced $= E_i - E_f$

$$= \frac{Q^2}{2}\left[\frac{1}{4\pi\varepsilon_0 R} - \frac{1}{2 \times 4\pi\varepsilon_0 R}\right]$$

$$= \frac{1}{4\pi\varepsilon_0} \cdot \frac{Q^2}{2R}\left[1 - \frac{1}{2}\right] \quad *$$

$$= \frac{1}{4\pi\varepsilon_0} \cdot \frac{Q^2}{4R}$$

21. **Ans: 3**

Solution: Resistance of galvanometer,
$G = 50\Omega$
Full scale current, $i_g = 0.05$
$A = 2.97 \times 10^{-2}$ cm^2
$= 2.97 \times 10^{-2} \times 10^{-4}$ m^2
$= 2.97 \times 10^{-6}$ m^2
$i = 5$ A
$\rho = 5 \times 10^{-7}\Omega - \text{m}$
Required resistance to convert the galvanometer into ammeter,
$R = \frac{i_g G}{i - i_g} = \frac{0.05 \times 50}{5 - 0.05} = \frac{25}{4.95}$
$\rho \frac{l}{A} = \frac{50}{99}$
$l = \frac{50}{99} \times \frac{A}{\rho}$
$= \frac{50}{99} \times \frac{2.97 \times 10^{-6}}{5 \times 10^{-7}}$
$= \frac{50}{99} \times \frac{29.7}{5}$
$= 10 \times 0.3$
$= 3$ m

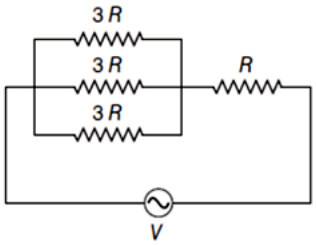
22. **Ans: 2**

Solution: Using Gauss's law, we have
$\oint \vec{E} \cdot d\vec{A} = \frac{1}{\epsilon_0}\int(\rho dv) = \frac{1}{\epsilon_0}\int_0^R kr^a \times 4\pi r^2 dr$
or $E \times 4\pi R^2 = \left(\frac{4\pi k}{\epsilon_0}\right)\frac{R^{(a+3)}}{(a+3)}$
$\therefore \quad E_1 = \frac{kR^{(a+1)}}{\epsilon_0(a+3)}$
For $r = \frac{R}{2} \cdot E_2 = \frac{k\left(\frac{R}{2}\right)^{a+1}}{\epsilon_0(a+3)}$
Given, $E_2 = \frac{E_1}{8}$
or $\frac{k\left(\frac{R}{2}\right)^{a+1}}{\epsilon_0(a+3)} = \frac{1}{8}\frac{kR^{(a+1)}}{\epsilon_0(a+3)}$
$\therefore \quad \frac{1}{a+1} = \frac{1}{8}$
or $a = 2.$

23. **Ans: 150**

Solution: Let resistance of 300 W bulb is R. Hence, $\frac{V^2}{R} = 300$ W.
$\therefore$ Resistance of 100 W bulb should be $3R$

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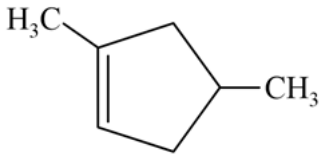
So, we have
$\therefore$ Equivalent resistance is $R_{\text{eq}} = 2R$
Hence, power $_{eq} = \frac{V^2}{(2R)}$
$= \frac{1}{2} \times \frac{V^2}{R} = \frac{300}{2} = 150$ W

24. **Ans: 6**

Solution: Due to cylinder: $E_1 = \frac{\rho R^2}{\varepsilon_0 \cdot 4R}$.
Due to sphere: $E_2 = \frac{1}{4\pi\varepsilon_0} \frac{\frac{4}{3}\pi\frac{R^3}{8}\rho}{(2R)^2}$.
Net field, $E_1 - E_2 = \frac{\rho R}{4\varepsilon_0} - \frac{\rho R}{4\varepsilon_0 \times 24} = \frac{23\rho R}{4\times 24\varepsilon_0} = \frac{23\rho R}{16K\varepsilon_0} \Rightarrow K = 6$

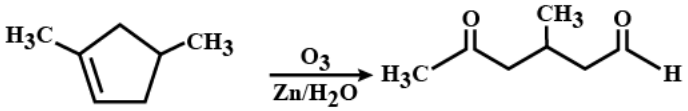
25. **Ans: 4**

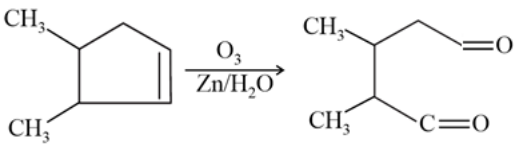
Solution: Capacitance of capacitors $C_1, C_3, C_4, C_5 = 4\mu$F each and capacitance of capacitor $C_2 = 10\mu$F. If a battery is applied across A and B, the points b and c will be at the same potential (since $C_1 = C_4 = C_3 = C_5 = 4\mu$F). Therefore no charge flows through C_2. We have the capacitors C_1 and C_5 in series. Therefore their equivalent capacitance,
$C' = \frac{C_1 \times C_5}{C_1 + C_5} = \frac{4\times 4}{4+4} = 2\mu$F
Similarly, C_4 and C_3 are in series. Therefore their equivalent capacitance,
$C'' = \frac{C_3 \times C_4}{C_3 + C_4} = \frac{4\times 4}{4+4} = 2\mu$F.
Now C' and C'' are in parallel. Therefore effective capacitance between A and B
$= C' + C'' = 2 + 2 = 4\mu$F.

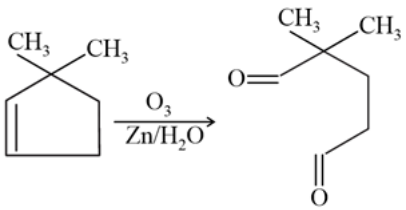


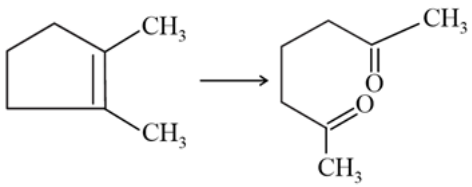
26. **Ans: (a)**

Solution:









All these three options will give the different products, only the option A will give the desired product by reductive ozonolysis.

27. **Ans: (a)** C_2H_5OH and C_2H_5ONa

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Solution: $\underset{\text{Ethanol}}{2C_2H_5OH} + 2Na \longrightarrow \underset{\text{Sod. ethoxide}}{2C_2H_5ONa} + H_2\uparrow$

$\underset{(A)}{}$ $\underset{(B)}{}$

$2C_2H_5OH \xrightarrow[\text{conc. } H_2SO_4]{} \Rightarrow \underset{\text{Diethyl ether}}{C_2H_5O \cdot C_2H_5}$

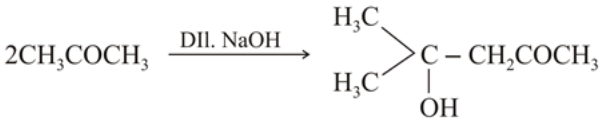
Thus, (A) is ethanol and (B) is sodium ethoxide.

28. **Ans: (a)** $2CH_3COCH_3 \xrightarrow{\text{Dil.NaOH}} CH_3COHCH_3CH_2COCH_3$

Solution:

Aldehyde and ketone having α-hydrogen atom undergo aldol condensation in presence of dilute base to give β −hydroxy aldehydes or ketones. Aldol condensation reaction results in the formation of a carbon-carbon bond between enol(or enolate) molecule with aldehyde or ketone. Acetone enolate is the nucleophile and another molecule of acetone is electrophile in this reaction.

The addition step without subsequent dehydration is an aldol reaction.
Acetone has α-hydrogen atom, hence it will give aldol condensation reaction.



β-hydroxy ketone

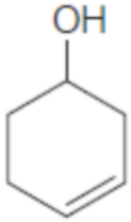
This β-hydroxy ketone further undergoes condensation reaction to form α, β unsaturated ketone.

29. **Ans: (c)** Hex-2 −ene-1 −ol

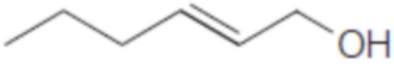
Solution:



(a) Hex3 −ene-1 −ol



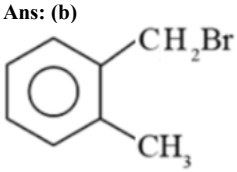
(b) cyclohex 3 −ene-1 −ol



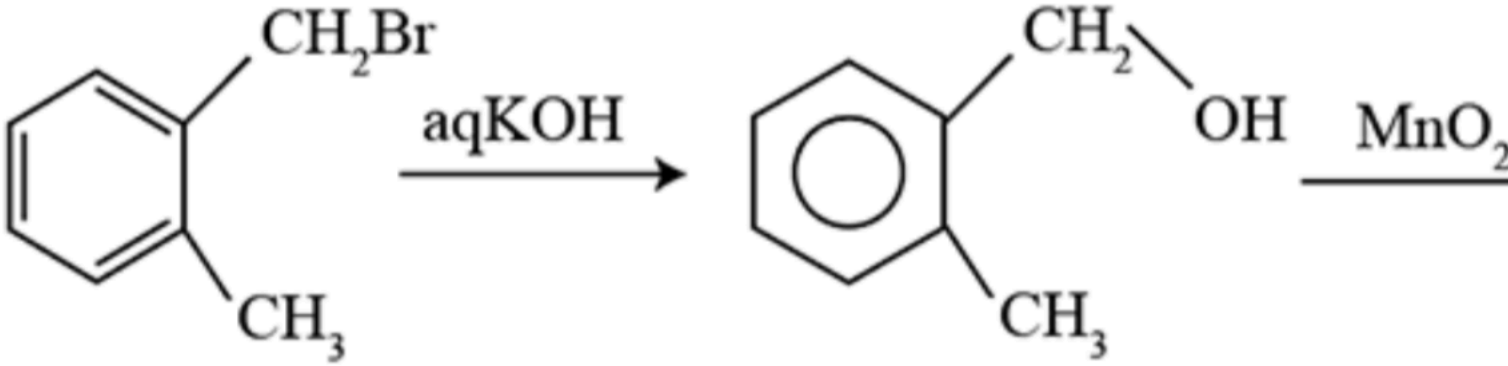
(c) Hex 2 −ene-1 −ol
This is allylic alcohol. Hence, it will undergo oxidation.

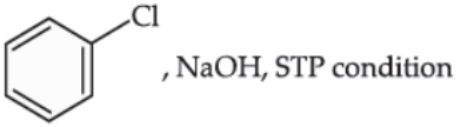


30. **Ans: (b)**



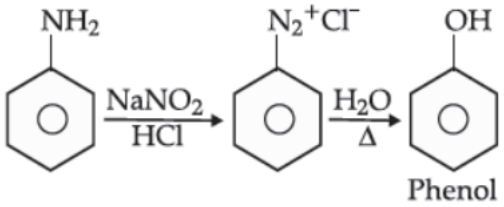
Solution:



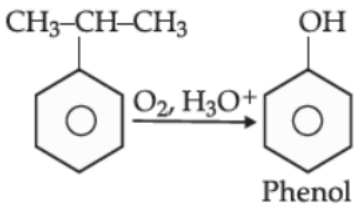


31. **Ans: (c)**

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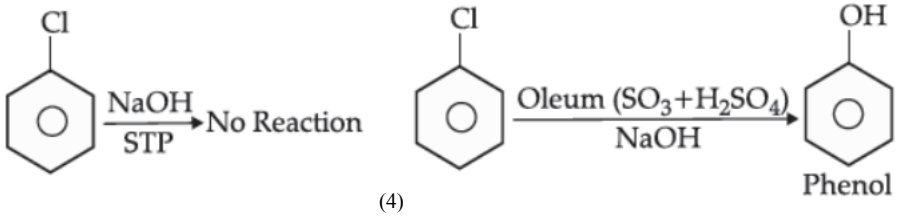


Solution: (1)



(2)

(3)



(4)

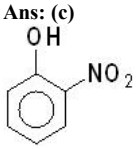
32. **Ans: (d)** CH_3CH_2OH

Solution: Key Idea (i) PBr_3 is a halogenating agent, ie, converts $-OH$ group into $-Br$. (ii) Alc. KOH is a dehydrohalogenating agent. (iii) H_2SO_4 and H_2O converts an olefin into alcohol. $\underset{\text{ethanol}}{C_2H_5OH} \xrightarrow{PBr_3} C_2H_5Br \xrightarrow[-KBr\ -H_2O]{\text{Alc.KOH}} CH_2=CH_2 \xrightarrow{H_2SO_4} CH_3-CH_2OSO_3H \xrightarrow[-H_2SO_4]{H_2O/\Delta} \underset{\text{ethanol}}{CH_3CH_2OH}$

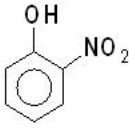
33. **Ans: (c)** $NaBH_4$

Solution: $CH_2=CH-CHO \xrightarrow{NaBH_4} CH_2=CH-CH_2OH$
$NaBH_4$ is selective reagent for the reduction of only carbonyl group.

34. **Ans: (c)**

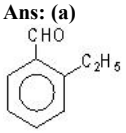


Solution: Due to internal hydrogen bonding, hydrogen of $-OH$ is strongly bonded. So it can not behave as an acid and can not react with sodium bicarbonate. Ortho - nitro phenol is non polar and $NaHCO_3$ is polar. Like dissolves like.



is not soluble in $NaHCO_3$

35. **Ans: (a)**



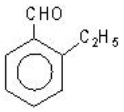
Solution: 1. X forms 2, 4-DNP derivatives, it shows that it is a carbonyl compound (C=0).

2. It reduces Tollen's reagent, it shows that it has an aldehyde group.

3. It undergoes Cannizaro reaction, that also shows the presence of an aldehyde having no α-hydrogen.

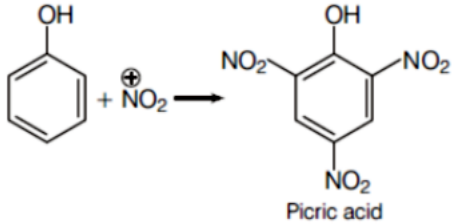
4. On vigorous oxidation, it produces 1, 2-benzenedicarboxylic acid. It shows that groups are present at 1,2-position on benzene ring.

Thus, the correct structure of the compound X is



36. **Ans: (d)** 7 , acidic

Solution: Phenol undergo electrophilic substitution reaction on reaction with a mixture of conc. HNO_3 and conc H_2SO_4 (known as nitrating mixture) This reaction produces nitro compound and on successive nitration it produces trinitrophenol.
conc. \mathrm{HNO}_{3}+ conc. \mathrm{H}_{2} \mathrm{SO}_{4} \longrightarrow \mathrm{NO}_{2}^{+}(electrophile)

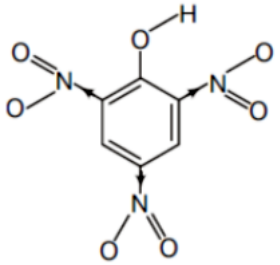


Molecular formula =\mathrm{C}_{6} \mathrm{H}_{3} \mathrm{~N}_{3} \mathrm{O}_{7}
Degree of unsaturation
=(C+1)-\frac{H}{2}-\frac{X}{2}+\frac{N}{2}
=(6+1)-\frac{3}{2}+\frac{3}{2}=7-0=7

u/wrotethatass

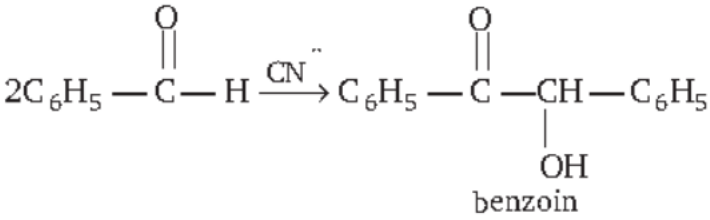
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Nature of compound The compound is acidic in nature due to presence of three strong electron withdrawing groups \left(\mathrm{NO}_{2}\right).

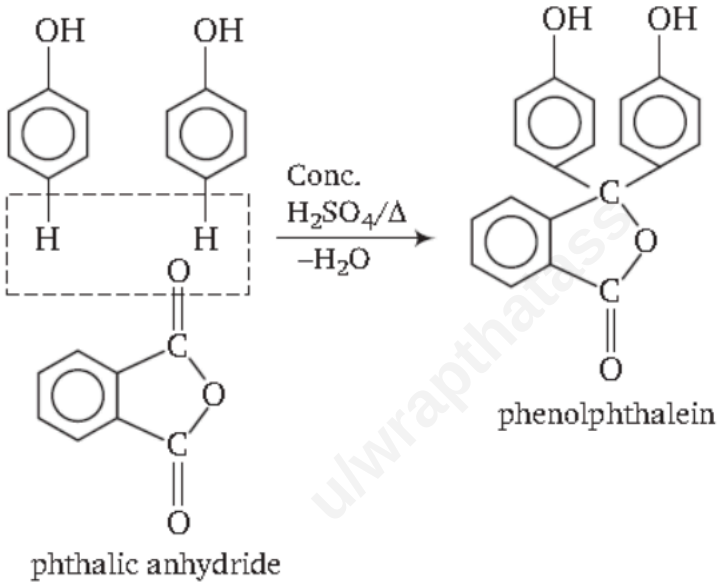


37. **Ans: (d)** (d)

Solution: (A) Benzoin condensation When an ethanolic solution of benzaldehyde is heated with strong alkali like \mathrm{KCN} or \mathrm{NaCN}, we get benzoin.

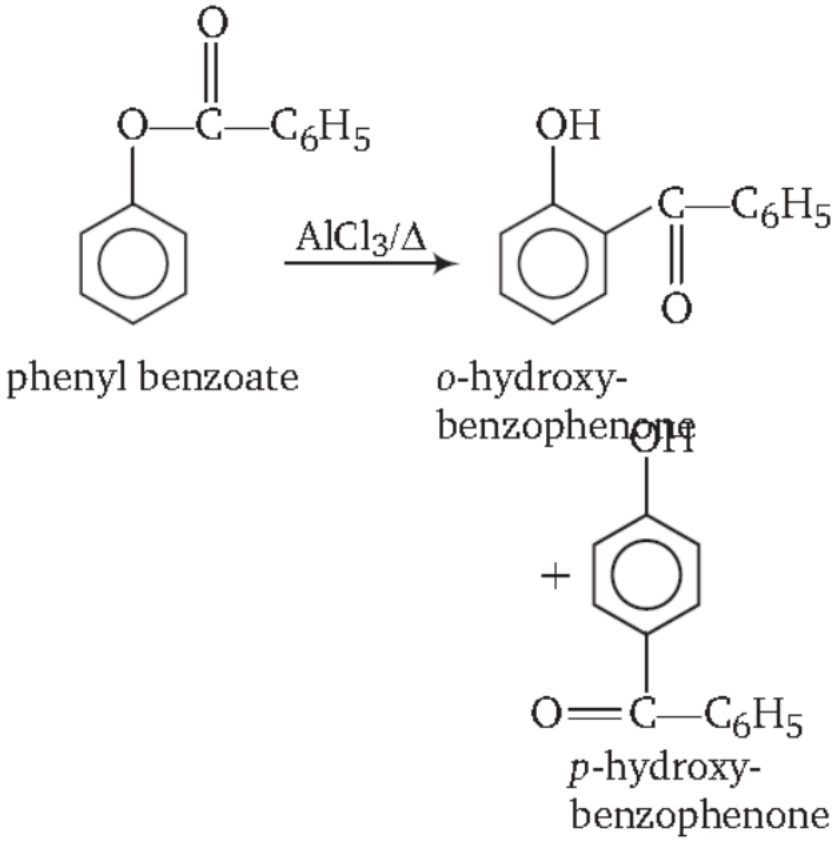


(B) Formation of phenolphthalein When phenol is reated with phthalic anhydride in the presence of concentrated \mathrm{H}_2 \mathrm{SO}_4, it gives phenolphthalein, an indicator.



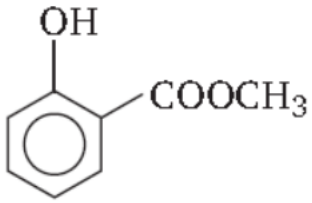
(C) Fries rearrangement Phenyl benzoate when heated with anhydrous \mathrm{AlCl}_3 in the presence of inert solvent gives ortho-and para-hydroxybenzophenone. In this rearrangement, there is a benzoyl group migration from the phenolic

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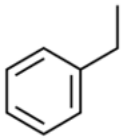


oxygen to an ortho-and para-position.

(D) Methylsalicylate



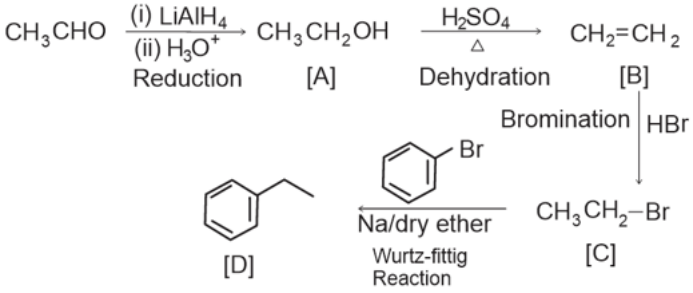
\text {(A chief constituent of oil of wintergreen)}

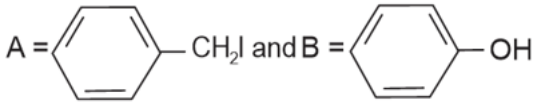


38. **Ans: (b)**

Solution:

is a strong reducing agent. It will reduce acetaldehyde to primary alcohol (Ethanol). Then ethanol on dehydration gives ethene. The formed alkene on bromination gives alkyl halide which undergo Wurtz-fittig reaction in the presence of sodium/dry ether. The complete reaction is given below:



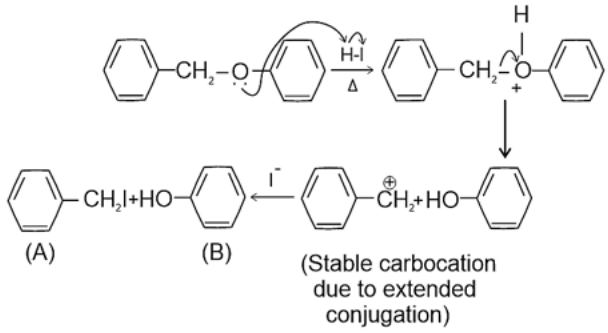


39. **Ans: (d)**

Solution:

Ether when treated with produces both benzyl iodide or phenol. Under acidic conditions (i.e., with a small amount of HI), the ether can undergo an acid-catalyzed cleavage to form phenol and benzyl iodide. In this reaction, the acts as a catalyst, and the protonation of the ether oxygen atom makes it a good leaving group, which then departs to form benzyl carbocation. The carbocation can then react with the iodide ion to form benzyl iodide, while the protonated oxygen atom can undergo deprotonation and form phenol.

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40. **Ans: (b)** Iodoform test

    Solution: $\mathrm{CH}_3 \mathrm{CHO}$ and $\mathrm{C}_6 \mathrm{H}_5 \mathrm{CH}_2 \mathrm{CHO}$ both being aliphatic aldehydes react with Tollen's reagent, Fehling solution and Benedict solution. So, these reagents cannot be used to distinguish them. $\mathrm{CH}_3 \mathrm{CHO}$ reacts with NaOH and I_2 to give yellow crystals of iodoform while $\mathrm{C}_6 \mathrm{H}_5 \mathrm{CH}_2 \mathrm{CHO}$ does not react with it. $\begin{aligned} \mathrm{CH}_3 \mathrm{CHO}+3 \mathrm{I}_2+ & 4 \mathrm{NaOH} \longrightarrow \mathrm{CHI}_3 +\mathrm{HCOONa}+3 \mathrm{NaI}+3 \mathrm{H}_2 \mathrm{O} \\ \mathrm{C}_6 \mathrm{H}_5 \mathrm{CH}_2 \mathrm{CHO}+\mathrm{I}_2 & +\mathrm{NaOH} \longrightarrow \text { No reaction } \end{aligned}$ Thus, $\mathrm{CH}_3 \mathrm{CHO}$ and $\mathrm{C}_6 \mathrm{H}_5 \mathrm{CH}_2 \mathrm{CHO}$ can be distinguished by iodoform test.

41. **Ans: (d)** $\begin{array}{ll} \qquad \quad \mathrm{C}_2 \mathrm{H}_5 \\ \qquad \quad \mid \\ \mathrm{CH}_3-\mathrm{C}(\mathrm{OH}) \mathrm{C}_6 \mathrm{H}_5 \end{array}$

    Solution: $\mathrm{CH}_3 \mathrm{COOH}+\mathrm{PCl}_5 \longrightarrow \underset{(a)}{\mathrm{CH}_3 \mathrm{COCl}} \xrightarrow[\mathrm{AlCl}_3]{\mathrm{CH}_6} \underset{(b)}{\mathrm{C}_6 \mathrm{H}_5 \mathrm{COCH}_3} \xrightarrow[\text { ether }]{\mathrm{C}_2 \mathrm{H}_5 \mathrm{MgBr}} \underset{(c)}{\begin{array}{ll} \qquad \quad \mathrm{C}_2 \mathrm{H}_5 \\ \qquad \quad \mid \\ \mathrm{C}_6 \mathrm{H}_5 \mathrm{C}-\mathrm{CH}_3 \\ \qquad \quad \mid \\ \qquad \quad \mathrm{OH} \end{array}}$

42. **Ans: (b)** Statement I is correct but Statement II is incorrect.

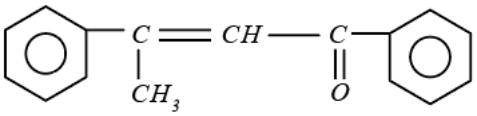
    Solution:

    The presence of the Nitro group in the ortho or para position decreases the electron density in the bond. As a result, it is easier to lose a proton from mono substituted nitro phenol Hence, ortho and para nitrophenols are stronger acids than phenol.

    The acidic nature of nitrophenols increases in the order: meta ortho para.

    Meta position shows only -I effect. Hence, it is less acidic than the para and ortho isomers.

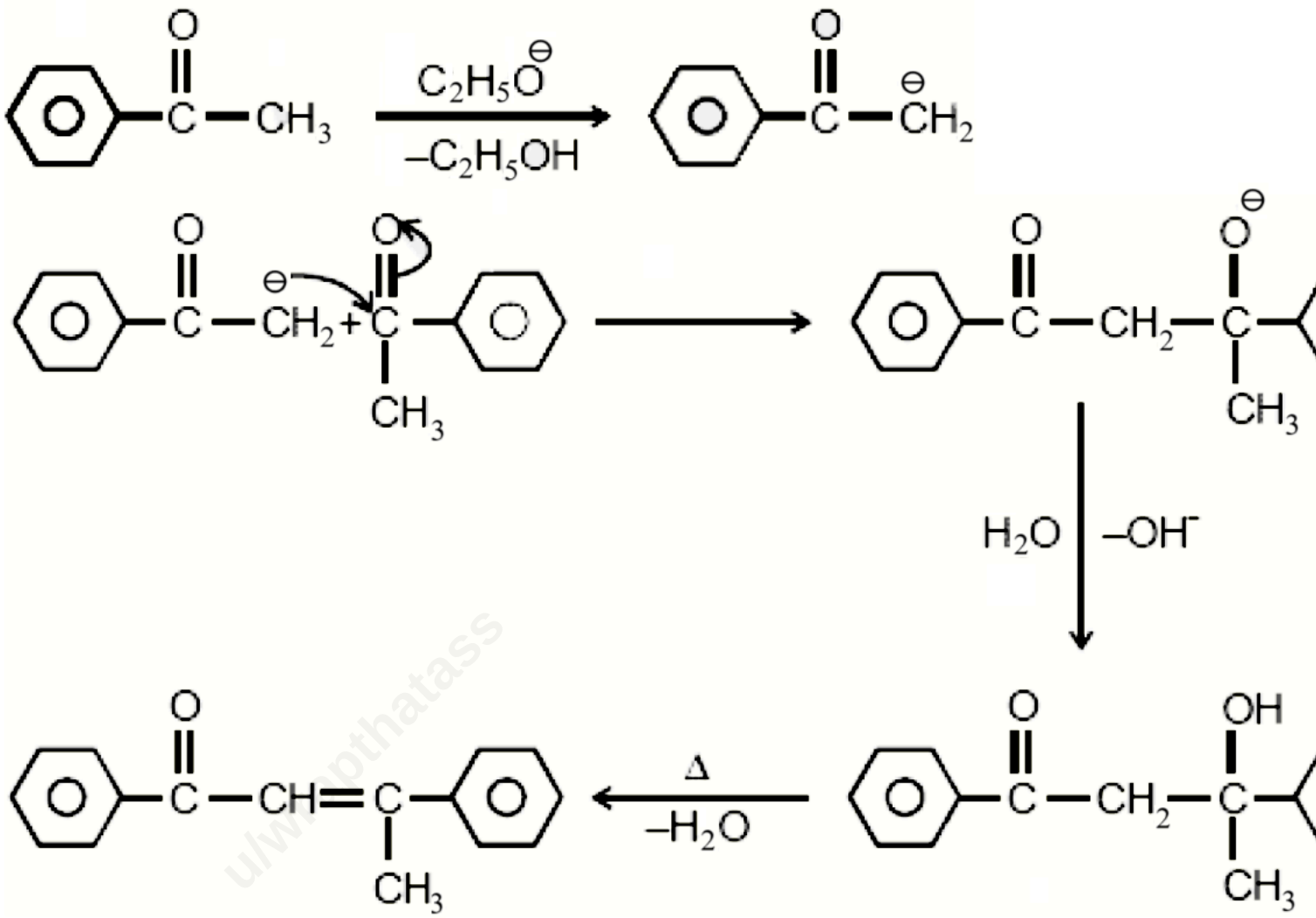
    Between para and ortho nitrophenols, para is more acidic due to the presence of an intramolecular hydrogen bonding that locks hydrogen of bond in ortho isomer.

43. **Ans: (b)** 

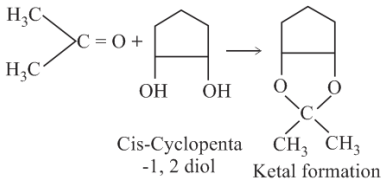
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Solution: It show condensation reaction



44. **Ans: (a)** Acetone

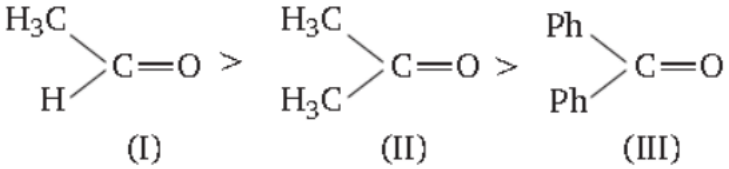


Solution:

While with trans cyclopenta -1,2 diol it will not form any product

45. **Ans: (d)** I > II > III

Solution: Since alkyl group has +I-effect and aryl group has +R-effect, thus greater the number of alkyl and aryl groups attached to the carbonyl group, its reactivity towards nucleophilic addition reaction. Secondly, as the steric crowding on carbonyl group increases, the reactivity decreases accordingly. \therefore Correct reactivity order for reaction

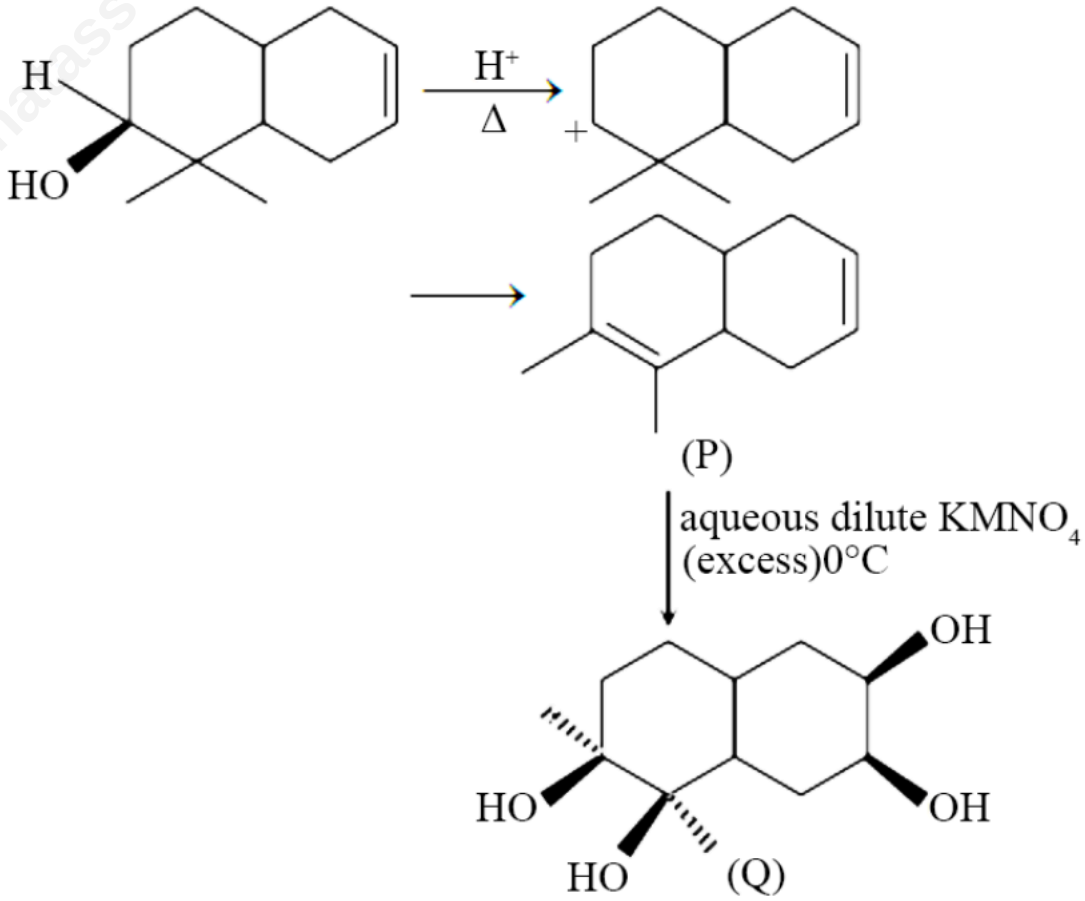


with \mathrm{PhMgBr} is

46. **Ans: 4**

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Solution:



47. **Ans: 4**

Solution: Alcohols (formula \mathrm{R}-\mathrm{CHOH}- \mathrm{CH}_3) gives iodoform test. Following isomers give positive iodoform test:

(I) $CH_3 - CH_2 - CH_2 - CH_2 - \underset{\underset{OH}{|}}{CH} - CH_3$

2-hexanol

(II) $CH_3 - CH_2 - \underset{\underset{CH_3}{|}}{CH} - \underset{\underset{OH}{|}}{CH} - CH_3$

3-methyl-2-pentanol

(III) $CH_3 - \underset{\underset{CH_3}{|}}{CH} - CH_2 - \underset{\underset{OH}{|}}{CH} - CH_3$

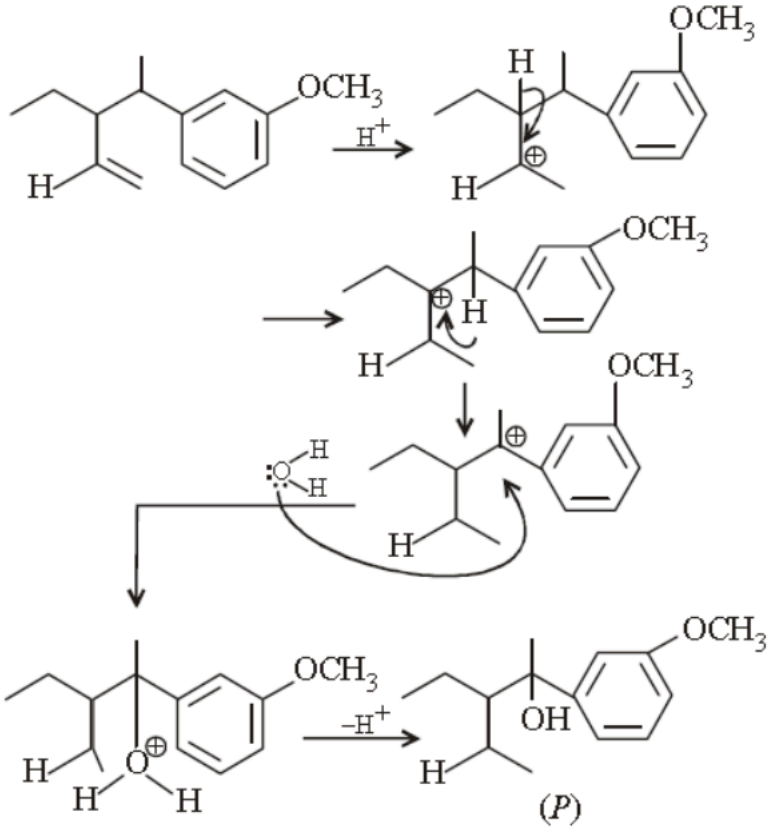
4-methyl-2-pentanol

(IV) $CH_3 - \underset{\underset{CH_3OH}{|}}{\overset{\overset{CH_3}{|}}{C}} - CH - CH_3$

3, 4-dimethyl-2-butanol

48. **Ans: 2**

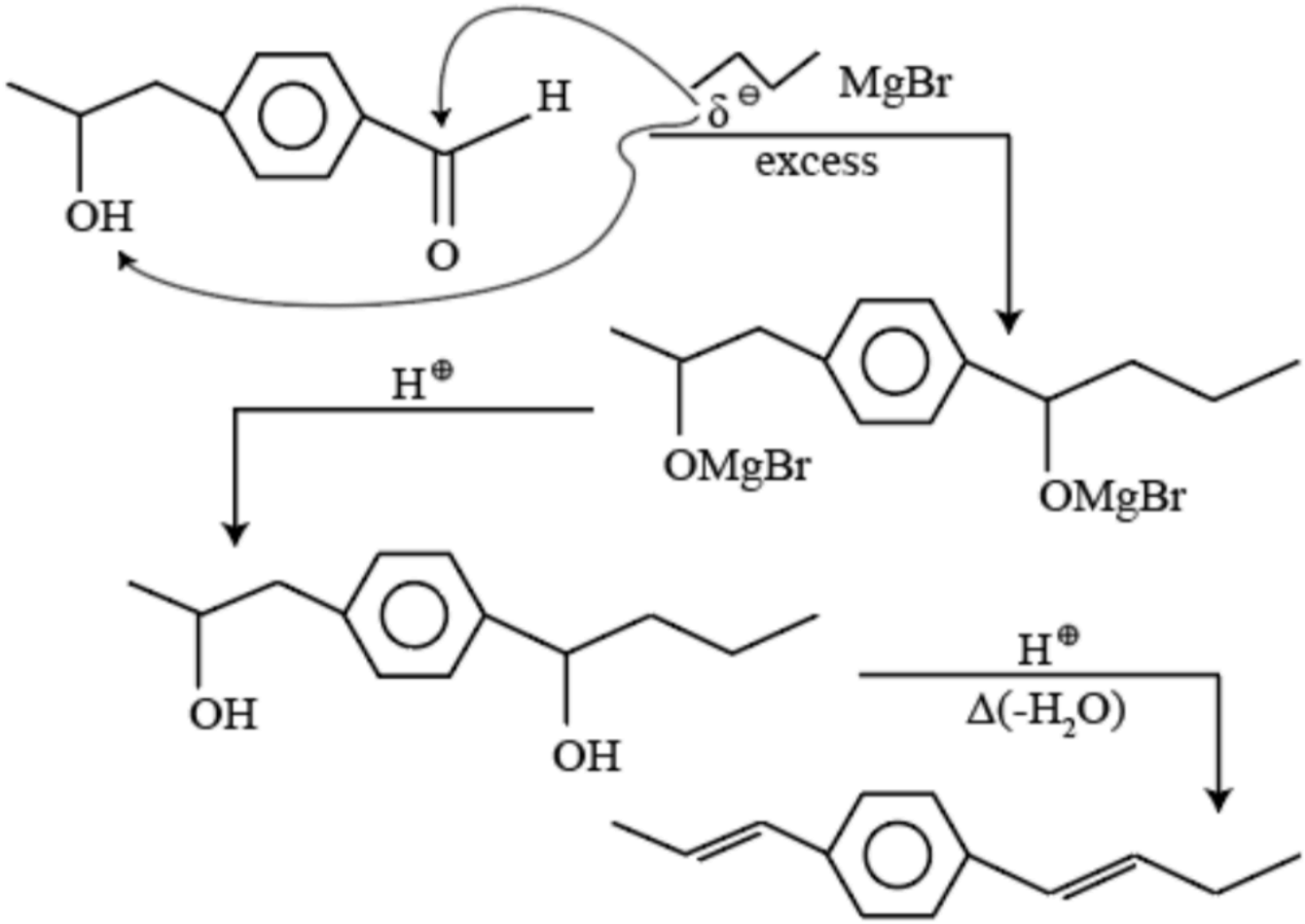
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Solution:

49. **Ans: 4**

Solution:



Here the product 'A' has two double bonds with different groups so it has four geometrical isomers.

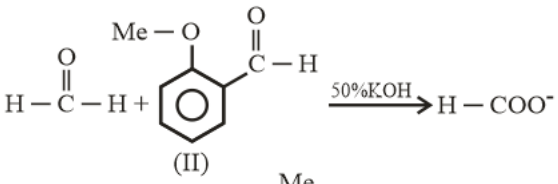
50. **Ans: 2**

Solution:

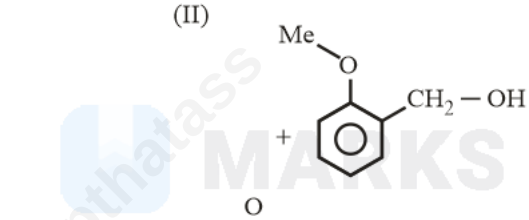
Aldehyde without give Cannizaro reaction.

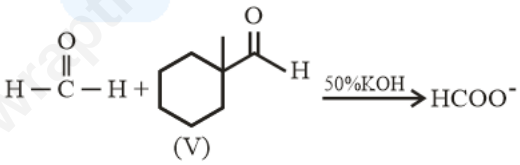
In Cannizaro reaction alcohol and carboxylic acid salt is formed.

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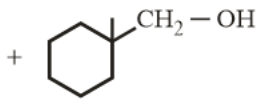


(II)





(V)



51. **Ans: (c)**

Solution:

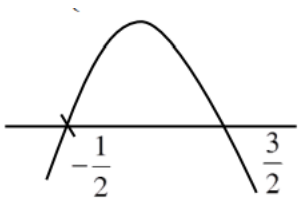
We know that, and

Hence,

And, using product rule, we get

For

Hence, the graph of the quadratic is



We know that, if then for those value of is increasing.

is increasing in

For

As

So, in complete domain, is increasing in

52. **Ans: (b)** values of

Solution:

(as is not possible and is true)

This is non-differentiable at and when

Hence, has exactly three points of non-differentiability.

53. **Ans: (a)** \mathrm{f}(\mathrm{x}) has local maximum at \mathrm{x}=1

Solution:
\begin{aligned} &f(x)=(x-1)^{2}(x-2)^{3} e^{x} \\ &\Rightarrow \mathrm{f}^{\prime}(\mathrm{x})=(\mathrm{x}-1)(\mathrm{x}-2)^{2}\left(\mathrm{x}^{2}+2 \mathrm{x}-5\right) \mathrm{e}^{\mathrm{x}} \end{aligned}
f(x) has local maximum at x=1, since f^{\prime}(x) changes from positive to negative.
\mathrm{f}(\mathrm{x}) has a point of inflexion at \mathrm{x}=2.

54. **Ans: (d)** \frac{2}{3}(1-a)

Solution: \lim _{x \rightarrow 2} \frac{a^{x}+a^{3-x}-\left(a^{2}+a\right)}{a^{3-x}-a^{x / 2}} \lim _{x \rightarrow 2} \frac{\left(a^{x}-a\right)\left(a^{x / 2}-a\right)\left(a^{x / 2}+a\right)}{\left(a-a^{x / 2}\right)\left(a^{2}+a^{x}+a^{x / 2} \cdot a\right)} \lim _{x \rightarrow 2}-\frac{\left(a^{x}-a\right)\left(a^{x / 2}+a\right)}{a^{2}+a^{x}+a^{x / 2} \cdot a} =\frac{2}{3}(1-a)

55. **Ans: (d)** 7

Solution: 1 \leq x^3 \leq 8 discontinuous at x^3=2,3,4,5,6,7. Hence non differentiable f(1)=-2 f(1+h)=-2 continuous and diff. at \begin{aligned} & \mathrm{x}=1 \\ & f(8)=(-2)^{8^3} \& f(2-\mathrm{h}) \\ & =\lim _{\mathrm{h} \rightarrow 0}(-2)^{[(8-\mathrm{h})]}=(-2)^7 \end{aligned} Non diff. for \mathrm{x}^3=8 Total points of non - differentiablity are - 7

56. **Ans: (c)**

Solution:

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\begin{aligned} & \text {when } x=0 \\ & m=0-0+2020=2020 \\ & \text {when } x=1 \\ & m=1-3+2020=2018\end{aligned}

For is strictly increasing

\begin{aligned} & \text {when } x=2 \\ & M=8-6+2020=2022\end{aligned}

57. **Ans: (d)** is continuous at but is not differentiable at

Solution:

Hence, Hence h^{\prime}(x) is continuous at x=0
So
Now,

Hence, is non-derivable at .

58. **Ans: (a)** 10 !

Solution: \lim _{x \rightarrow \infty}\left[\frac{1^{\frac{1}{x}}+2^{\frac{1}{x}}+3^{\frac{1}{x}}+\ldots+10^{\frac{1}{x}}}{10}\right]^{10 x} on
=\lim _{y \rightarrow 0^{+}}\left[\frac{1^{y}+2^{y}+3^{y}+\ldots+10^{y}}{10}\right]^{\frac{10}{y}}~\left\{ \text{where, } y=\frac{1}{x}\right\}
=e^{y \rightarrow 0^{+}} \frac{10}{y}\left[\frac{1^{y}+2^{y}+3 y_{+\ldots}+10^{y}}{10}-1\right]
\left.=e^{\lim _{y \rightarrow 0^{+}}\left[\frac{1^{y}+2^{y}+3^{y}+\ldots+10^{y}-10}{y}\right]}{y}\right]
=e^{\lim _{y \rightarrow 0^{+}}\left[\frac{\left(1^{y}-1\right)}{y}+\frac{\left(2^{y}-1\right)}{y}+\frac{\left(3^{y}-1\right)}{y}+\ldots+\frac{\left(10^{y}-1\right)}{y}\right]}
=e^{(\log 1+\log 2+\log 3+\ldots+\log 10)}
=e^{\log (1 \cdot 2 \cdot 3 \ldots . .10)}=10 !

59. **Ans: (a)** \mathrm{f}(\mathrm{x}) is always a constant function

Solution: \mathrm{f}^{\prime}(\mathrm{x}) \leq 2 \mathrm{f}(\mathrm{x})
\mathrm{f}^{\prime}(\mathrm{x}) \mathrm{e}^{-2 \mathrm{x}} \leq 2 \mathrm{f}(\mathrm{x}) \mathrm{e}^{-2 \mathrm{x}}
\frac{\mathrm{d}}{\mathrm{dx}}\left(\mathrm{f}(\mathrm{x}) \mathrm{e}^{-2 \mathrm{n}}\right) \leq 0
\mathrm{~g}(\mathrm{x})=\mathrm{f}(\mathrm{x}) \mathrm{e}^{-2 \mathrm{x}} is non-Increasing function
\mathrm{x} \geq 0
\mathrm{~g}(\mathrm{x}) \leq \mathrm{g}(0)
\mathrm{f}(\mathrm{x}) \mathrm{e}^{-2 \mathrm{x}} \leq \mathrm{f}(0) \mathrm{e}^{-0}
\mathrm{f}(\mathrm{x}) \mathrm{e}^{-2 \mathrm{x}} \leq 0
\mathrm{f}(\mathrm{x}) \leq 0 \quad but given \mathrm{f}(\mathrm{x}) is not negative
\therefore \mathrm{f}(\mathrm{x})=0 \quad Constant function

60. **Ans: (c)**

Solution:

Since, , where is an integer

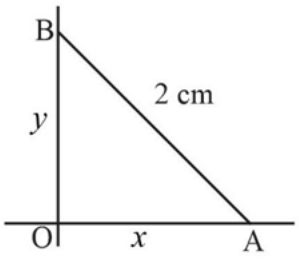
and

Therefore,

Since fractional part of any number lies between and .

Therefore,

61. **Ans: (d)**



Solution:

,
Now,
differentiating w.r.t. both side
when

62. **Ans: (c)**

Solution:

Let
so,

As,

is increasing monotonically,
Now, and
So, this function must intersect -axis at one point.

So, number of solution.

63. **Ans: (d)**

Solution:

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is discontinuous at

is discontinuous at

Now

Hence is continuous at

Therefore, the points of discontinuity are and

64. **Ans: (d)**

Solution:

Given equation:

Suppose,

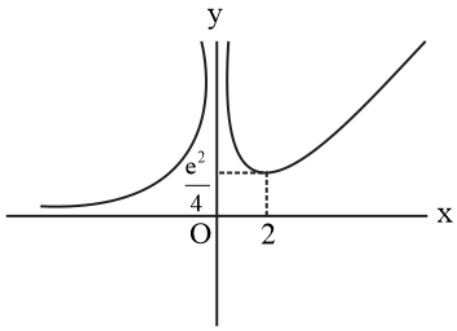
For critical points,

or

Now, is a minima as and .

Now,

The graph of the function :



For three distinct solutions,

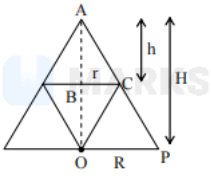
as in that case it will definitely have three points of intersection with the curve, so,

.

65. **Ans: (a)** approaches $+\infty$

Solution: $\lim _{x \rightarrow 0}\left\{\frac{1}{x^{2}}+\frac{(2013)^{x}}{e^{x}-1}-\frac{1}{e^{x}-1}\right\}$
$=\lim _{x \rightarrow 0}\left\{\frac{1}{x^{2}}+\frac{(2013)^{x}-1}{e^{x}-1}\right\}$
$=\lim _{x \rightarrow 0}\left\{\frac{1}{x^{2}}+\frac{(2013)^{x}-1}{x} \cdot \frac{x}{e^{x}-1}\right\}$
$=\lim _{x \rightarrow 0} \frac{1}{x^{2}}+\lim _{x \rightarrow 0} \frac{(2013)^{x}-1}{x} \cdot \lim _{x \rightarrow 0} \frac{x}{e^{x}-1}$
$=+\infty+\log (2013) \cdot 1$
$=+\infty$

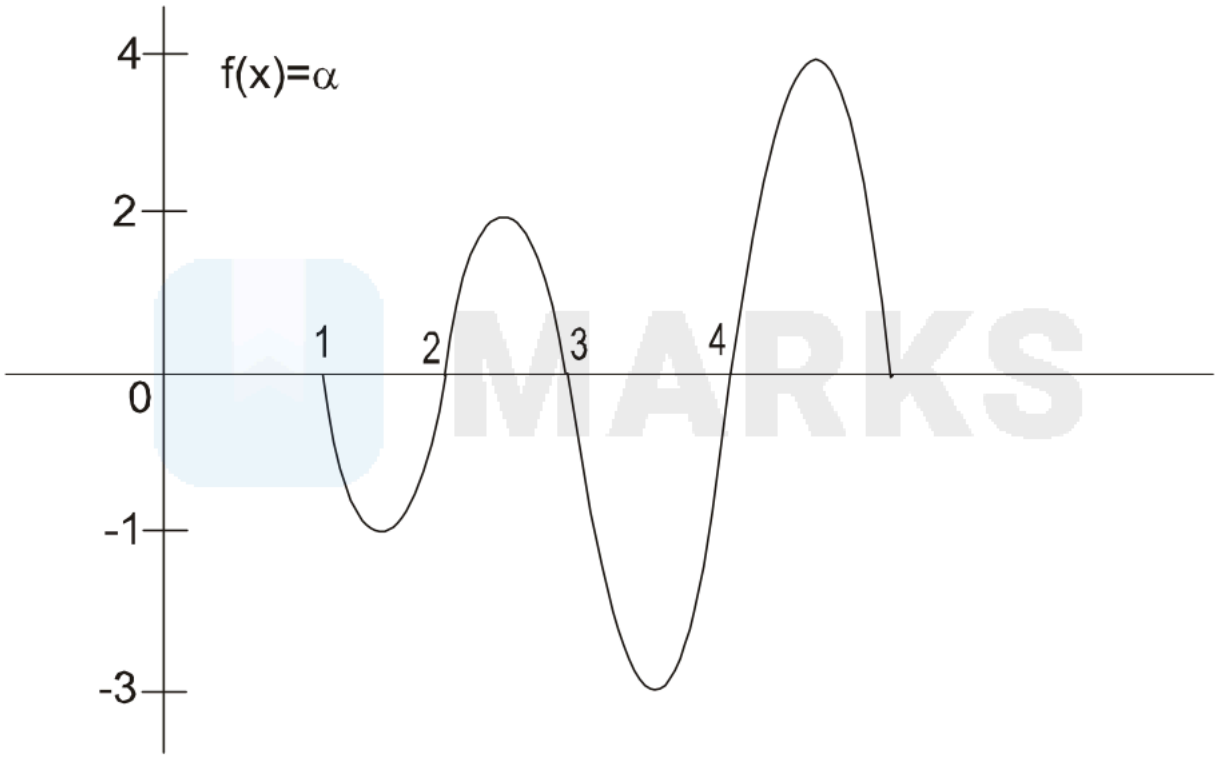
66. **Ans: (c)** $\frac{4}{27}$



Solution:
$\triangle \mathrm{ABC}$ and $\triangle \mathrm{AOP}$ are similar
$\begin{array}{l} \frac{\mathrm{h}}{\mathrm{r}}=\frac{\mathrm{H}}{\mathrm{R}} \quad \Rightarrow \mathrm{h}=\frac{\mathrm{rH}}{\mathrm{R}} \\ \mathrm{V}_{2}=\frac{1}{3} \pi \mathrm{r}^{2}(\mathrm{H}-\mathrm{h})=\frac{\pi}{3} \mathrm{r}^{2} \mathrm{H}\left(1-\frac{\mathrm{r}}{\mathrm{R}}\right)=\frac{\pi \mathrm{H}}{3}\left(\mathrm{r}^{3}-\frac{\mathrm{r}^{3}}{\mathrm{R}}\right) \\ \frac{\mathrm{d} \mathrm{V}_{2}}{\mathrm{dr}}=2 \mathrm{r}-\frac{3 \mathrm{r}^{2}}{\mathrm{R}}=0 \quad \mathrm{r}=\frac{2 \mathrm{R}}{3} \\ \mathrm{~V}_{2 \max }=\frac{4 \pi \mathrm{R}^{2} \mathrm{H}}{81} \end{array}$
$\begin{array}{l} \mathrm{V}_{1}=\frac{\pi \mathrm{R}^{2} \mathrm{H}}{3} \\ \therefore \frac{\mathrm{V}_{2}}{\mathrm{~V}_{1}}=\frac{4}{27} \end{array}$

67. **Ans: (d)** For each real α, the equation $f(x)-\alpha=0$ has infinitely many roots.

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Solution: f(x)=[x] \sin \pi x

68. **Ans: (b)** (0,1)\cup(1,4)

Solution: f(x)=\left(a^{2}=-3 a+2\right)\left(\cos ^{2} x / 4-\sin ^{2} x / 4\right)+(a-1) x+\sin 1
\Rightarrow \mathrm{f}(\mathrm{x})=(\mathrm{a}-1)(\mathrm{a}-2) \cos \mathrm{x} / 2+(\mathrm{a}-1) \mathrm{x}+\sin 1
\Rightarrow \mathrm{f}^{\prime}(\mathrm{x})=-\frac{1}{2}(\mathrm{a}-1)(\mathrm{a}-2) \sin \frac{\mathrm{x}}{2}+(\mathrm{a}-1)
\Rightarrow \mathrm{f}^{\prime}(\mathrm{x})=(\mathrm{a}-1)\left[1-\frac{(\mathrm{a}-2)}{2} \sin \frac{\mathrm{x}}{2}\right]
If f(x) does not possess critical points, then f^{\prime}(x) = 0 for any x \in R
\Rightarrow(a-1)\left[1-\frac{(a-2)}{2} \sin \frac{x}{2}\right] = 0 for any x \in R
\Rightarrow \mathrm{a}=1 and 1-\left(\frac{\mathrm{a}-2}{2}\right) \sin \frac{\mathrm{x}}{2}=0
must not have any solution in \mathrm{R}.
\Rightarrow \mathrm{a}=1 and \sin \frac{\mathrm{x}}{2}=\frac{2}{\mathrm{a}-2} is not solvable in \mathrm{R}.
\Rightarrow \mathrm{a} = 1 and \left|\frac{2}{\mathrm{a}-2}\right|>1\left[\right. For \mathrm{a}=2, \mathrm{f}(\mathrm{x})=\mathrm{x}+\sin 1 \therefore
\mathrm{f}^{\prime}(\mathrm{x})=1
\Rightarrow \mathrm{a}=1 and 0<\mathrm{a}<4 \Rightarrow \mathrm{a} \in(0,1) \cup(1,4).

69. **Ans: (a)** 2

Solution: \begin{aligned} & \lim _{a \rightarrow \infty}\left(\frac{\left(a^2+8 a+3\right)-\left(a^2+4 a+2\right)}{\sqrt{a^2+8 a+3}+\sqrt{a^2+4 a+2}}\right)=2 \\ & \therefore \quad \tan ^2 \alpha+\cot ^2 \alpha=2 \Rightarrow \tan ^2 \alpha=1 \\ & \Rightarrow \sin ^2 \alpha=\cos ^2 \alpha=\frac{1}{2} \\ & \sum_{n=1}^{\infty}\left(\sin ^{2 n} \alpha+\cos ^{2 n} \alpha\right)=\sum_{n=1}^{\infty}\left[\left(\frac{1}{2}\right)^n+\left(\frac{1}{2}\right)^n\right]=2\end{aligned}

70. **Ans: (b)** at exactly two values of x

Solution: f(x)=\left\{\begin{array}{ll}\frac{x+5}{x-2}, & x \neq 2 \\ 1, & x=2\end{array}\right. f(x) is discontinuous at x=2 f(x)=2 \Rightarrow \frac{x+5}{x-2}=2
\begin{array}{l} \Rightarrow x+5=2 x-4 \\ \Rightarrow x=9 \end{array}

71. **Ans: 2**

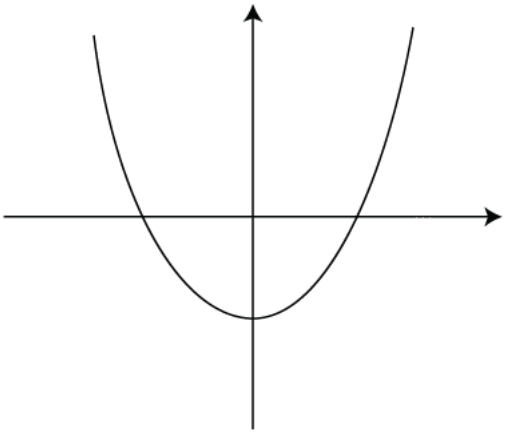
Solution:

Given that

72. **Ans: 2**

Solution: Let, f(x)=x^{4}+4 x^{3}+12 x^{2}+7 x-3
\therefore f^{\prime}(x)=4 x^{3}+12 x^{2}+24 x+7
f^{\prime \prime}(x)=12\left(x^{2}+2 x+2\right)
i.e. f^{\prime \prime}(x) > 0, \quad \forall x \in R
Also, f(0)=-3

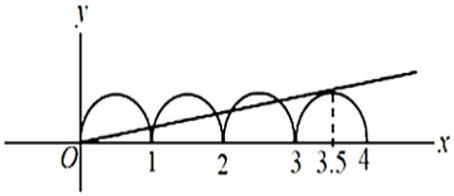
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Hence, f(x)=0 has 2 real roots

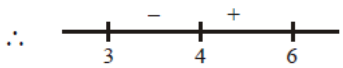
73. **Ans: 4**

Solution:



Thus, for the maximum points of non-differentiability, graphs of and must intersect at maximum number of points which occurs when

74. **Ans: 1**



minimum at x = 4

maximum at x = 3 or 6.

Solution: f(x)=\int_3^\pi t\left(t^2-7 t+12\right) d t, f^{\prime}(x)=x(x-3)(x-4) \begin{aligned} & \therefore f(3)=0, f(6)=\int_3^6\left(t^3-7 t^2-12 t\right) d t \\ & =\left(\frac{t^4}{4}-\frac{7 t^3}{3}+\frac{12 t^2}{2}\right)_3^6=\frac{99}{4}\end{aligned} f(x)=\int_3^4\left(t^3-7 t^2+12 t\right) d t=-\frac{7}{12} \therefore \quad 3 a+\frac{b}{9}=1

75. **Ans: 0**

Solution:

We know that,

For,

And, for

We have

Now, Left hand Limit, at

And Right hand limit, at

And,

Therefore, at at

Hence, the function is continuous at

For, So, is continuous for all

And for is a polynomial function of degree which is always continuous for all Real value of Therefore is continuous for all

Hence is continuous for all

And,

Therefore, Left hand derivative at Right hand derivative at

Therefore, is differentiable at

For, for all values of So, is differentiable for all

For, is a polynomial function of degree which is differentiable for all values .

Hence, is continuous and differentiable for all So, number of points in the interval where is not differentiable is Zero.

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